MIL-C-43528A (GL)
30 September 1985
SUPERSEDING
MIL-C-43528 (GL)
2 August 1967

MILITARY SPECIFICATION

CONVEYOR, ROLLER, GRAVITY, 1000 LB/FT

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of roller conveyor.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank
MIL-W-22248 - Weldments, Aluminum and Aluminum Alloy
MIL-S-22473 - Sealing, Locking, and Retaining Compounds: (Single-Component)
MIL-C-52950 - Crates, Wood, Open and Covered

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3910

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

11-1-589 - Conveyor, Roller Gravity, Airdrop, Aluminum, 1000 Lb/Ft Capacity - Assembly
11-1-590 - Frame - Assembly and Details
11-1-591 - Roller - Assembly and Details
11-1-592 - Coupling - Assembly and Details
11-1-593 - Shafts, Roller

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 20.1 - Safety Standard for Conveyors and Related Equipment

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials. All materials shall conform to the documents subsidiary to the drawings listed in section 2 and to the requirements specified herein. Recycled material may be used (see 6.4).

* 3.3 Construction. Construction and assembly shall be as specified herein and on the drawings listed in section 2. The conveyor shall conform to the applicable requirements of ASME B 20.1.

3.3.1 Weldment. The aluminum frame weldment, as specified in Drawing 11-1-590, shall be fabricated by the inert-gas, tungsten-arc welding process and shall conform to the requirements applicable to class 3 weldments as specified by MIL-W-22248, except that the requirement for identification marking shall not apply. In lieu thereof, the part number and weldment serial number shall be applied in accordance with Drawing 11-1-590. Inspection marking requirements shall remain as specified.

3.3.2 Roller assemblies. The shell ends shall accommodate the bearings (see 6.5) with the interference fit specified on the detail drawing. The bearings shall be free of lubricant. Prior to the insertion of the bearing into the shell, the entire surface area of contact shall be coated with sealing compound conforming to grade AA of MIL-S-22473. Exposed excess compound shall be removed after insertion.

3.3.2.1 Roller location. Unless otherwise specified, rollers shall be installed at the hexagonal hole locations at each end of the frame and at alternate hole locations thereafter.

3.4 Performance. The conveyor shall show no evidence of failure when tested as specified in 4.5.

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* 3.5 Identification marking. An identification plate conforming to type III, composition A class 2 of MIL-P-514 shall be the size and place in the location shown on Drawing 11-1-589.

* 3.6 Workmanship. Completed conveyor shall be free from dents, scratches, tool marks, splinters, sharp edges, welding voids, cracks and all evidence of rough handling or damage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), the gravity roller conveyor shall be examined as specified in 4.4.2 and 4.4.3. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, materials and components shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

4.4.2 Intermediate inspection. Examination shall be made of aluminum frame weldments for visual defects, proper dimensions, and fluorescent penetrant. The sample unit shall be one aluminum frame weldment, as specified in Drawing 11-1-590. The inspection level shall be S-3. Any defect found shall be cause for rejection of the lot represented by the sample unit.

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4.4.2.1 Visual and dimensional examination. Characteristics to be examined, defects to be scored, and acceptability standard shall be as prescribed for class 3 weldments by MIL-W-22248.

4.4.2.2 Fluorescent penetrant examination. Two lateral support welds of each sample, selected at random, shall be examined in accordance with all requirements for methods, procedures and acceptability standards prescribed for class 3 weldments by MIL-W-22248.

* 4.4.3 End item inspection. Completed conveyors shall be examined for visual defects and proper dimensions. The sample unit shall be one complete conveyor, as specified in Drawing 11-1-589.

4.4.3.1 Examination of visual characteristics. Characteristics to be examined and classification of defects shall be as listed in table I. The inspection level shall be II and the acceptable quality level (AQL) shall be 1.5 for major defects, and 2.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

Examine	Defect	Category	
		Major	Minor
Construction and workmanship	Any component missing, cracked, fractured, misaligned, or otherwise damaged, except as otherwise classified herein	101	
Hardware (screws, nuts, knobs and pins)	Missing, wrong type or size, stripped, or loose	102	
	Parts not finished as specified	103	
Roller assemblies	Shell dented, cracked, broken; or bearing not securely retained	104	
	Dirt, sealing compound, or other foreign matter in bearing preventing free rolling action	105	

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TABLE I. Classification of visual defects - Continued

Examine	Defect	Category	
		Major	Minor
Identification plate and marking	Plate missing or not located as specified, incorrect or illegible		201
	Serial number or part number not as specified on Drawing 11-1-590		202

4.4.3.2 Examination of dimensions. Examination shall be made of dimensional characteristics pertaining to the assembled end item (i.e., overall frame dimensions, roller position intervals, location of lateral support and spacers and location of couplings). The inspection level shall be S-4 and the AQL shall be 1.5 defects per hundred units. Each deviation from specified dimensions and tolerances shall constitute a defect.

* 4.4.4 Packaging inspection. An examination shall be made to determine compliance with the packing and marking requirements specified in section 5 of this document. Examination shall be made of shipping containers fully packaged. The lot shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. Characteristics to be examined and the defects applicable thereto shall be as specified in table II. The inspection level shall be S-1 and the AQL shall be 2.5 defects per hundred units.

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TABLE II. Packaging inspection

Examine	Defect
Marking	Omitted, incorrect, not legible, or of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, defective, or not as specified
Workmanship	Contents not snug fit in container (loose); inadequate blocking, bracing, strapping, or nailing

4.5 Performance test. The sample unit for this test shall be one complete conveyor, Drawing 11-1-589. The conveyor shall be supported along the side rails at intervals of approximately 30 inches between centers. The bearing surface of each support shall provide approximately 4 square inches of surface contact with the side rails. A load of two thousand (2000) pounds shall be evenly distributed on a rigid bearing platform faced with 1/8-inch thick aluminum sheet. The bearing platform shall be constructed to dimensions approximately 20 inches wide by 30 inches long. The aluminum sheet facing shall span six rollers, center to center, providing the test load support of five rollers minimum, (approximately two (2) linear feet of conveyor). The load shall be cycled (rolled from one end of the conveyor to the other and return) five (5) times at a velocity of 175 (plus 10 or minus 5) feet per minute. The load shall be removed and the conveyor examined for defects indicated in table III. The examination shall be conducted at inspection level S-2. Any defect shall be cause for rejection of the lot represented by the sample.

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TABLE III. Performance inspection

Examine	Defect
Welds	Crack, break or other deformation
Roller assemblies	Dent, bend, bearing displaced from retaining dimples or not otherwise secure in shell, bound bearing, not otherwise free running roller, bent twisted or otherwise deformed shaft
Side rails, lateral supports, spacers and coupling	Twist, bend, crack, break or deformation

5. PACKAGING

5.1 Packing. Packing shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Level A packing. Six conveyors, with coupling assemblies fully retracted and secured into the frame assemblies, shall be stacked flat in a snug-fitting shipping container conforming to type II, style A of MIL-C-52950. Each alternate conveyor in the stack shall be offset to permit nesting of the rollers within the container. The applicable anchoring, closing and strapping requirements specified in the appendix of MIL-C-52950 shall apply. Each container shall be fitted with skids fabricated from nominal 3 by 4 inch lumber laid flat extending the full width of the container and positioned under each bottom cross member. The end of each skid shall be beveled one-half its thickness at a 45 degree angle and notched to provide clearance for the girthwise strapping. Each skid shall be secured to the container by nailing through the bottom cross member into the skid. The nail points shall not protrude through the bottom surface of the skid. The nails shall be arranged in two rows in a staggered pattern, with spacing between nails in each row not more than 6 inches. Each row of nails shall be approximately 3/4 inch in from the edges of the skid and the nailing pattern shall begin and end approximately 1-1/2 inches from the beveled end of each skid.

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★ 5.1.2 Commercial packing. Conveyors shall be packed in accordance with ASTM D 3951.

★ 5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

★ 5.2.1 Special marking. Each container shall be marked in accordance with the applicable marking requirements specified in MIL-C-52950.

6. NOTES

6.1 Intended use. The high capacity roller conveyor is intended for use in support of airborne operations and other load conveying applications involving loads not exceeding 1000 pounds per linear foot when the conveyor is supported at 30-inch intervals along the side rails. The conveyor is modular with provision for coupling.

★ 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Number and location of rollers if other than specified (see 3.3.2.1).
d. Selection of applicable level of packing (see 5.1).

★ 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first article should consist of the unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

★ 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

★ 6.5 Roller assembly bearings. Bearings furnished as Part Number 242-R1Z by Conveyor Logic, 6900 Industrial Park Drive, Dutton, MI 49316, conform to the requirements of this document.

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6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only and the Government assumes no liability whatsoever for inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43528A	Conveyor, Roller Gravity, 1000 Lb/Ft

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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